

M.Tech Research Literature Survey

Project Title: AI-Based Digital Brain Twin for Predicting Neurological Disease Progression Using MRI, EEG and Clinical Data

1. Required Literature Survey Table

S.NO	TITLE OF THE PAPER	NAME OF JOURNAL / CONFERENCE	AUTHORS	VOLUME, ISSUE, YEAR	METHODOLOGY USED	APPLICATION
1	A Unified Approach to Interpreting Model Predictions	NeurIPS (NIPS) 2017 / arXiv: 1705.07874	Scott M. Lundberg, Su-In Lee	31st Conf. on NeurIPS, 2017	Additive feature attribution framework unifying LIME, DeepLIFT, Layer-wise Relevance Propagation, Shapley regression/sampling values, and QII; derives SHAP values as the unique solution satisfying local accuracy, missingness, and consistency; introduces Kernel SHAP (weighted linear regression), Max SHAP, and Deep SHAP (DeepLIFT-style backpropagation).	Explainable AI — post-hoc feature-importance attribution for complex/black-box ML models (decision trees, deep nets).
2	Digital twins for health: a scoping review	npj Digital Medicine	Katsoulakis, Wang, Wu, Shahriyari, Fletcher, Liu, Achenie, Liu, Jackson, Xiao, Syeda-Mahmood, Tuli, Deng	(2024) 7:77	PRISMA-informed scoping review; PubMed search (2016–2023) for "digital twin" + "health"; bibliometric/ word-cloud analysis of titles.	Broad taxonomy of Digital Twin for Health (DT4H): hospital management, device design, drug discovery, biomanufacturing, surgical planning, clinical trials, personalized medicine, wellness.
3	Digital Twins in Health Care: Ethical Implications of an Emerging Engineering Paradigm	Frontiers in Genetics	Koen Bruynseels, Filippo Santoni de Sio, Jeroen van den Hoven	Vol. 9, Article 31 (2018)	Conceptual/philosophical analogical analysis comparing industrial Digital Twin engineering paradigm to personalized-medicine "virtual patients"; no computational model.	Ethical/conceptual analysis of DT impact on definitions of health, disease, therapy, prevention, and enhancement.
4	Tutorial: Brain Network Analysis with Graph Neural Networks	Tutorial proposal (BrainGB-associated; IEEE BigData/KDD-style tutorial)	Hejie Cui, Xuan Kan, Xiaoxiao Li, Lifang He, Liang Zhan, Ying Guo, Carl Yang	Not Reported (tutorial document; underlying BrainGB benchmark published in IEEE TMI, 2022)	Modularized survey of GNN design space for brain networks: node feature construction, message-passing mechanisms, attention-enhanced message passing, pooling strategies; also covers multimodal, dynamic, generative, and interpretable GNN extensions.	Brain network analysis (functional/structural connectomes from fMRI/DTI) for neurological/psychiatric disorder classification.
5	Multimodal Machine Learning: A Survey and Taxonomy	arXiv:1705.09406 (later IEEE TPAMI)	Tadas Baltrušaitis, Chaitanya Ahuja, Louis-Philippe Morency	Not Reported in this version (arXiv preprint, 2017)	Taxonomy survey across 5 challenges: Representation (joint/coordinated), Translation (example-based/ generative), Alignment (explicit/implicit), Fusion (model-agnostic/model-based: early/late/hybrid, MKL, graphical models, neural nets), Co-learning (parallel/non-parallel/ hybrid data).	General multimodal ML framework applicable to AVSR, image/video captioning, VQA, emotion recognition, cross-modal retrieval.
6	Neural Digital Twins: Toward Next-Generation Brain-Computer Interfaces	IEEE (submitted/ preprint)	Mohammad Mahdi Habibi Bina, Sepideh Baghernezhad, Mohammad Reza Daliri, Mohammad Hassan Moradi	2026 (search conducted Nov 8, 2025; not yet formally published)	PRISMA systematic review (606→23 papers); proposes 5-layer "BCI-DT" architecture (Physical, Virtual, Digital Thread, Analysis, Feedback layers); deep learning (CNN, LSTM, transformer/POYO), Kalman-filter data assimilation.	Next-generation BCIs — continuous adaptation, personalized neural modeling, predictive cortical simulation, multimodal EEG/fMRI/ECoG fusion for neuroprosthetics.

S.NO	TITLE OF THE PAPER	NAME OF JOURNAL / CONFERENCE	AUTHORS	VOLUME, ISSUE, YEAR	METHODOLOGY USED	APPLICATION
7	Swin Transformer: Hierarchical Vision Transformer using Shifted Windows	ICCV 2021	Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, Baining Guo	pp. 10012–10022 (2021)	Hierarchical Vision Transformer with shifted-window multi-head self-attention (W-MSA/SW-MSA), patch merging, relative position bias; linear complexity in image size.	General-purpose vision backbone: image classification, object detection, semantic segmentation.
8	U-Net: Convolutional Networks for Biomedical Image Segmentation	MICCAI 2015 / arXiv: 1505.04597	Olaf Ronneberger, Philipp Fischer, Thomas Brox	2015	U-shaped fully convolutional network — contracting path (3×3 conv + ReLU + 2×2 max-pool) and symmetric expanding path (up-conv + skip-concatenation with cropped features); pixel-wise softmax with weighted cross-entropy; elastic-deformation data augmentation; SGD, momentum 0.99.	Biomedical image segmentation: neuronal structures in EM stacks, cell segmentation/tracking in light microscopy.

2. Observation (Results, Merits & Demerits)

1. SHAP

- *Results*: Kernel SHAP matches true Shapley values with far fewer model evaluations than Shapley sampling/LIME; user studies (30 and 52 participants) show SHAP agrees with human explanations more than LIME/DeepLIFT; Deep SHAP raises the change-in-log-odds metric on MNIST masking (~45) versus original DeepLIFT (~32).
- *Merits*: Unifies six prior explanation methods under one theoretical framework; unique solution satisfying local accuracy, missingness, consistency; both model-agnostic (Kernel SHAP) and model-specific (Max SHAP, Deep SHAP) variants.
- *Demerits*: Exact computation is combinatorially intractable, requiring approximation; several fast variants assume feature independence, often violated in real (e.g., clinical) data; small user-study sample sizes; no direct handling of highly correlated multimodal features (relevant if used for MRI/EEG feature attribution).

2. Digital Twins for Health (Scoping Review)

- *Results*: 220 DT4H papers identified via PubMed (2016–2023), 85 used for synthesis; publication counts rose from 2 (2016) to 321 (2023) with rising DT4H:DT ratio.
- *Merits*: Broad 8-category taxonomy of DT4H applications; catalogues major international DT research consortia (SDTC, PRIMAGE, DT4H.org, etc.); clarifies DT maturity levels (representation → model → shadow → intelligent twin).
- *Demerits*: No quantitative meta-analysis (scoping review only); DT4H described as still largely conceptual/early-stage; explicitly flags data interoperability, privacy, bias, and infrastructure as unresolved; no brain-specific digital twin architecture proposed.

3. Digital Twins in Health Care: Ethical Implications

- *Results*: N/A — conceptual paper. Identifies emergent categories such as the "asymptomatic ill" and blurring of therapy/enhancement/prevention boundaries under a DT paradigm.
- *Merits*: Rich ethical/philosophical grounding for discussing personalized "virtual patients"; anticipates governance, privacy, and equity issues relevant to any brain-twin deployment.
- *Demerits*: Purely theoretical — no technical methodology, dataset, or computational model; conclusions are speculative about future DT capability.

4. GNN Brain Network Tutorial

- *Results*: N/A (tutorial/survey); references outcomes from BrainGNN, BrainNetCNN, and the BrainGB benchmark package.
- *Merits*: Modular design-space breakdown (node features, message passing, attention, pooling) directly transferable to disease-progression GNNs on brain connectomes; provides ready-to-use BrainGB toolkit; covers multimodal and dynamic (temporal) brain-network GNNs — directly relevant to a digital brain twin.
- *Demerits*: Ethical constraints around human neuroimaging keep most brain-network datasets private, hurting reproducibility; deep GNNs need more labeled data than is typically available for rare neurological cohorts; not a primary research contribution (pedagogical framing).

5. Multimodal Machine Learning Survey

- *Results*: N/A — survey; synthesizes results/approaches from dozens of underlying papers rather than reporting original metrics.
- *Merits*: The five-challenge taxonomy (representation, translation, alignment, fusion, co-learning) is directly applicable to MRI+EEG+clinical fusion design decisions; distinguishes joint vs. coordinated representations and early/late/hybrid fusion trade-offs.
- *Demerits*: No original experiments; taxonomic categories can overlap; risks going stale quickly given the pace of multimodal deep learning (published 2017, pre-dating transformer-based fusion).

6. Neural Digital Twins for BCIs

- *Results*: PRISMA screening narrowed 606 records to 23 included studies; no new experimental results — a conceptual/architectural contribution (BCI-DT).
- *Merits*: Directly targets BCI limitations (latency, calibration drift, inter-subject variability, invasiveness) with a concrete 5-layer closed-loop architecture (Physical → Virtual → Digital Thread → Analysis → Feedback); reviews real-world neural-twin initiatives (Living Brain Project, Virtual Brain Twin, Neurotwin, Digital Twin Brain Simulator) with multimodal EEG/ECOG/fMRI integration — the closest existing analog to your proposed AI-based Digital Brain Twin.
- *Demerits*: No original computational model or experimental validation of the proposed BCI-DT framework itself; invasive BCI methods carry surgical risk; the field still lacks standardized evaluation metrics.

7. Swin Transformer

- *Results*: 87.3% top-1 (ImageNet-1K, Swin-L); 58.7 box AP / 51.1 mask AP (COCO test-dev); 53.5 mIoU (ADE20K val) — state-of-the-art at publication.
- *Merits*: Linear (not quadratic) computational complexity vs. image size; hierarchical multi-scale features compatible with FPN/U-Net-style decoders — well suited to 3D MRI segmentation/feature extraction;

shifted-window scheme gives ViT-level accuracy with CNN-like efficiency.

- *Demerits*: Window size is a manually chosen hyperparameter; best results still require large-scale pretraining (ImageNet-22K); implementation not as kernel-optimized as mature CNN backbones, so real-world latency gains are partly offset.

8. U-Net

- *Results*: ISBI EM-segmentation challenge — warping error 0.000353, rand error 0.0382 (best score at time of publication, beating Ciresan et al.'s sliding-window CNN at 0.000420/0.0504); PhC-U373 IOU 92.03%; DIC-HeLa IOU 77.56%.
- *Merits*: Learns from very few annotated images (20–35) via elastic-deformation augmentation; sub-second inference for 512×512 images; large margin over prior sliding-window CNN methods — foundational for MRI segmentation pipelines.
- *Demerits*: Unpadded convolutions shrink output relative to input (requires border handling/mirroring); original architecture is 2D only (3D MRI needs 3D U-Net extension); trained/evaluated on small, single-institution datasets, raising generalization concerns.

3. Associated Resource Identifiers & Alignment Links

PAPER	DATASET SOURCE & DESCRIPTION	PAPER LINK	DATASET LINK
SHAP	No dedicated dataset for core theory; empirical validation uses MNIST digit images and synthetic/decision-tree models; Amazon Mechanical Turk user study (n=30, n=52).	https://arxiv.org/abs/1705.07874	MNIST: http://yann.lecun.com/exdb/mnist/
DT4H Scoping Review	No public dataset used. Literature corpus: 220 PubMed-indexed papers screened, 85 synthesized.	https://doi.org/10.1038/s41746-024-01073-0	Not Provided
DT Ethics	No public dataset used. Framework/Conceptual study.	doi: 10.3389/fgene.2018.00031	Not Provided
GNN Brain Tutorial	No single dataset; discusses fMRI/DTI-derived structural & functional connectomes generally; BrainGB package supports multiple public brain-network datasets.	https://braingb.us	https://braingb.us
Multimodal ML Survey	No public dataset used. Survey/taxonomy paper.	https://arxiv.org/abs/1705.09406	Not Provided
Neural DT (BCI)	No public dataset used. PRISMA-reviewed corpus: 606 records (Scopus 388, PubMed 218) → 23 included.	Not Reported (preprint in submission to IEEE)	Not Provided
Swin Transformer	ImageNet-1K (1.28M train/50K val, 1000 classes); ImageNet-22K (14.2M images, 22K classes); COCO 2017 (118K train/5K val/20K test-dev); ADE20K (25K images, 150 categories). All public.	https://github.com/microsoft/Swin-Transformer	ImageNet: https://www.image-net.org/ • COCO: https://cocodataset.org/ • ADE20K: https://groups.csail.mit.edu/vision/datasets/ADE20K/
U-Net	ISBI 2012 EM Segmentation Challenge (30 training images, 512×512, Drosophila VNC, public); ISBI Cell Tracking Challenge 2015 — PhC-U373 (35 training images), DIC-HeLa (20 training images).	https://arxiv.org/abs/1505.04597	EM challenge: http://brainiac2.mit.edu/isbi_challenge/ • Cell tracking: http://www.codesolorzano.com/celltrackingchallenge/

4. Comparative Analysis

PAPER	STRENGTH	LIMITATION	RESEARCH GAP	HOW MY PROJECT IMPROVES IT
SHAP	Theoretically unique, consistent feature-attribution framework; strong human-alignment.	Assumes feature independence in fast approximations; not designed for multimodal, correlated clinical/imaging/EEG features.	No explainability tailored to fused MRI+EEG+clinical inputs; no longitudinal explanation of <i>disease progression</i> .	Apply SHAP/Deep SHAP (or an adapted multimodal variant) to explain each modality's contribution to progression predictions from the Digital Brain Twin, addressing the No Explainable AI gap.
DT4H Scoping Review	Comprehensive taxonomy of DT applications across healthcare.	No neuroscience-specific or brain-disease DT architecture; not systematic/quantitative.	No Digital Twin specific to brain disease progression combining imaging + electrophysiology + clinical data.	Proposes a concrete Brain-specific Digital Twin filling the gap this review identifies as still "early stage".
DT Ethics Paper	Deep conceptual grounding for personalization, consent, and equity issues.	Purely theoretical, no technical/computational contribution.	No operational framework translating ethical principles into an actual predictive DT system.	Embeds these ethical considerations (transparency, bias mitigation, personalized "normal" baselines) into the design of the proposed AI-based Brain Twin.
GNN Brain Tutorial	Modular, reusable GNN design space for brain connectomes; interpretable/dynamic/multimodal extensions already surveyed.	Only covers static/dynamic connectivity graphs, not integrated with imaging+EEG+clinical fusion or with a Digital Twin abstraction.	No fusion of GNN brain-network modeling with Digital Twin architecture or with longitudinal progression prediction.	Uses GNN brain-network representations as one modeling component inside the Digital Brain Twin, adding MRI+EEG+clinical fusion and progression prediction absent here.
Multimodal ML Survey	General-purpose taxonomy (representation/translation/alignment/fusion/co-learning) directly reusable for design choices.	No brain-specific or clinical case study; pre-transformer era (2017), missing modern fusion transformers.	No application of this taxonomy to a neurological Digital Twin combining MRI, EEG, and clinical data.	Directly instantiates the fusion/alignment/co-learning taxonomy for MRI+EEG+clinical data fusion within the Digital Brain Twin.
Neural DT (BCI)	Closest existing analog: 5-layer BCI-DT architecture; multimodal EEG/ECOG/fMRI integration; addresses BCI-specific limitations.	No original computational model/experimental validation; framework not extended to disease-progression prediction or explainability.	No Explainable AI, no disease-progression prediction (focus is BCI control, not diagnosis/prognosis).	Extends the BCI-DT layered architecture toward disease-progression prediction with explainable, longitudinal, multimodal MRI+EEG+clinical modeling.
Swin Transformer	State-of-the-art hierarchical vision backbone, linear complexity, strong for dense prediction (segmentation).	Not adapted for medical/3D imaging or multimodal fusion; needs large-scale pretraining.	No integration into a personalized, longitudinal neurological Digital Twin.	Could serve as the MRI feature-extraction backbone within the Digital Brain Twin, combined with EEG/clinical branches and explainability.
U-Net	Highly data-efficient segmentation architecture, strong baseline for biomedical imaging.	2D-only design in original form; single-modality (imaging only), no clinical/EEG fusion or longitudinal modeling.	No longitudinal, multimodal, or explainable extension of segmentation to disease-progression prediction.	Used as an MRI preprocessing/segmentation module feeding into the multimodal, explainable, longitudinal Digital Brain Twin.

5. Final Summary & Structural Synthesis

1. Common methodologies used: Across the corpus, three technical families recur: (a) deep learning backbones for perception — CNN/U-Net-style encoders for imaging and hierarchical/shifted-window Transformers (Swin) for dense vision tasks; (b) graph neural networks for structured brain-connectivity data; and (c) explainable AI (SHAP) for post-hoc interpretation of any of the above. Separately, the Digital Twin literature converges on a layered architecture — physical entity, virtual/digital model, and a bidirectional communication/digital-thread layer — increasingly augmented with AI/ML for adaptive, real-time personalization.

2. Most frequently used datasets: For the technical/ML papers: ImageNet-1K/22K, COCO, ADE20K (vision benchmarks), and the ISBI EM/cell-tracking challenges (biomedical segmentation). For the Digital Twin/health papers: no shared public datasets — most are literature reviews, conceptual papers, or reference proprietary/institutional initiatives (e.g., Living Brain Project, Virtual Brain Twin, PRIMAGE), reflecting the persistent lack of open neuro-digital-twin datasets.

3. Most common evaluation metrics: Top-1 accuracy, box/mask AP, mIoU, Dice/IOU, warping/rand/pixel error (vision & segmentation papers); for SHAP, agreement with human-assigned feature importance and sample efficiency; the DT/health/ethics/tutorial papers report no quantitative metrics, relying on qualitative synthesis (PRISMA counts, taxonomy coverage).

4. Current research trends: A clear trend toward (i) hierarchical, efficient Transformer backbones replacing pure CNNs for imaging; (ii) GNNs becoming the default tool for brain-connectivity analysis; (iii) explainability (SHAP-family methods) increasingly demanded alongside high-accuracy black-box models; and (iv) Digital Twin technology migrating from industrial/aerospace origins into healthcare and, most recently, into neuroscience/BCI as "Neural Digital Twins."

5. Major limitations across the literature: Persistent themes include (a) small, non-longitudinal, single-modality datasets in biomedical imaging; (b) lack of true multimodal (imaging + electrophysiology + clinical) fusion in existing brain-network or digital-twin work; (c) explainability methods not adapted for correlated multimodal biomedical features; (d) Digital Twin healthcare literature remaining largely conceptual, with few validated computational implementations; and (e) BCI-specific digital twins (the closest analog to the proposed project) not yet extended to general disease-progression prediction or explainability.

6. Unexplored research gaps: No reviewed paper combines all of the following: (i) an actual Digital Twin architecture, (ii) fused MRI + EEG + clinical data, (iii) explainable AI for feature-level interpretation, and (iv) longitudinal disease-progression prediction with clinical decision-support integration. The GNN-brain-network and multimodal-ML surveys supply the component techniques; the DT4H/ethics reviews supply the framework and governance considerations; the Neural-Digital-Twin/BCI paper supplies the closest architectural analog — but none unify these into a single predictive, explainable, multimodal neurological twin.

7. Why an AI-Based Digital Brain Twin is a meaningful and novel research direction: The literature shows mature, independent building blocks — high-accuracy segmentation/vision backbones (U-Net, Swin), brain-connectivity GNNs, rigorous explainability theory (SHAP), a well-articulated multimodal-fusion taxonomy, and an emerging Digital Twin paradigm already validated in general healthcare and initial BCI contexts — but no existing work integrates them into a unified, explainable, longitudinal Digital Brain Twin for neurological disease-progression prediction using MRI, EEG, and clinical data simultaneously. This positions the proposed M.Tech project at a genuine convergence point of five active but currently disconnected research threads, addressing a gap each individual paper implicitly or explicitly identifies.